

Muhammad Asad

Energy Data Analyst | Data Scientist

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Profile

Petroleum engineering graduate and data analyst currently pursuing a Master's in Energy Engineering at the University of Genoa. Builds statistically rigorous analysis, production-shaped machine learning pipelines, and full-stack data products, with a research background spanning peer-reviewed publications and international conference presentations. Comfortable across the full pipeline – from data collection and statistical analysis in Python/R to interactive dashboards (Power BI, Streamlit, Plotly) and full-stack web development (React, Next.js, FastAPI). Also active as a freelance web developer, designing and shipping production websites for clients end to end.

Skills

Data Visualization	Power BI, DAX, Power Query, Streamlit, Plotly (Express & Graph Objects), Tableau, interactive dashboard design
Programming & Data	Python (pandas, NumPy, scikit-learn), R, SQL, JavaScript/TypeScript, HTML5/CSS3
ML & Statistics	XGBoost, TensorFlow/Keras, SHAP, hypothesis testing, predictive modelling, ML pipelines for operational/research optimization
Web Development	React, Next.js, FastAPI, full-stack site design and deployment
Tools & Platforms	Docker, Kubernetes, PostgreSQL, Git/GitHub, MS Project, MS Excel/PowerPoint/Word
Research & Comm.	Technical writing, peer-reviewed publication, conference presentation, cross-disciplinary collaboration
Languages	English (Fluent), Urdu (Native), Italian (Basic)

Professional Experience

Research Analyst Feb 2024 – Present

Cloud Innovators Solution (CIS), Karachi

- Apply data analytics and ML pipelines to support operational optimization across multi-disciplinary projects.
- Design dashboards and data visualizations to communicate statistical findings to technical and business stakeholders.
- Conduct advanced statistical analysis and build structured project documentation/plans using MS Project.
- Prepare technical documentation and research reports for cross-functional teams.

Freelance Web Developer 2023 – Present

Self-employed, Remote

- Design and build client websites end to end – UI design, front-end development, hosting, and domain management – for clients including SmartPath AI and ProLayers.
- Delivered 7+ production websites spanning SaaS/consultancy platforms, e-commerce catalogs, and marketing sites.

Research Assistant May 2021 – Aug 2023

NED University of Engineering and Technology, Karachi

- Conducted experimental testing, data analysis, and literature review on shale swelling, drilling fluid rheology, and asphaltene precipitation.
- Implemented ML algorithms to predict fluid properties and stability parameters, translating experimental data into predictive models.
- Co-authored 5+ peer-reviewed papers published in *Chemical Papers* and *Arabian Journal for Science & Engineering*.

Data Analyst Intern Jun 2022 – Jul 2022

GlobalShala (Sponsored by Saint Louis University) – Remote

- Delivered audience-tailored data presentations, applying visual perception principles for effective, accessible communication of analytical findings.

Featured Projects

Energy Intelligence Platform 2026

FastAPI, PostgreSQL, Next.js/TypeScript, Docker Compose – Personal project

- Full-stack application over 25 years of European electricity-mix data: FastAPI + PostgreSQL backend, Next.js/TypeScript frontend, trend-based forecasting, containerized with Docker Compose and wired to GitHub Actions CI.

- The one project that goes all the way from raw data to a deployed product rather than stopping at a notebook or dashboard.

Credit Card Fraud Detection

2026

Python, scikit-learn, XGBoost, Streamlit – Personal project

- Analyzed real transaction data under extreme 0.17% class imbalance, comparing PR-AUC vs. ROC-AUC and supervised vs. unsupervised anomaly detection.
- Built a cost-based threshold optimizer that raised projected net fraud-prevention value by 74% over the naive 0.5 cutoff, shipped as an interactive Streamlit monitoring dashboard.

A/B Testing: Mobile Game Retention Analysis

2026

Python, pandas, SciPy – Personal project

- Ran a full statistical analysis of a real 90K-player experiment: a Sample Ratio Mismatch check before trusting any result, two-proportion z-tests cross-validated against 10,000-resample bootstrap confidence intervals.
- Simulated the effect of repeated significance testing ("peeking"), demonstrating a concrete false-positive scenario the real data would have produced under that practice.

Image Classification: CNN vs. Transfer Learning

2026

TensorFlow, Keras, Grad-CAM – Personal project

- Trained a CNN from scratch and compared it against MobileNetV2 transfer learning on real flower photographs.
- Ran an augmentation ablation study and used Grad-CAM to analyze misclassified images, translating model errors into visual, explainable diagnostics.

Oil Production Decline Curve Analysis

2026

Python, SciPy, Arps models – Personal project

- Applied classical petroleum-engineering decline curve analysis (Arps exponential/hyperbolic/harmonic models) to 60 years of real country-level oil production data.

SmartPath AI – Corporate & Healthcare Web Platform

Nov 2025 – Feb 2026

smpathai.com, healthcare.smpathai.com – Full-stack build & deployment

- Designed, developed, and deployed a corporate site and a separate healthcare-vertical platform end to end for a freelance client, owning the project from concept through production.
- Handled complete technical delivery: front-end development, hosting/infrastructure setup, and domain (DNS) management and registration.

Publications & Conference Contributions

- Co-authored 5+ peer-reviewed publications in *Chemical Papers* and *Arabian Journal for Science and Engineering*, covering shale stability, green drilling fluids, and AI-driven reservoir modeling.
- **SPE ADIPEC 2023** – *Investigating the Effect of Salt, Polymer and Magnetic Nanoparticles on Critical Drilling Parameters, Coupled with a Novel 3D Image Processing Technique.*
- **ACS Acknowledged Work:** *Carbon Capture and Storage (CCS) in Oil and Gas.*
- **Geothermal Rising Student Contest** (Abstract Selected): *Enhanced Geothermal Systems: Global Challenges & Strategies.*
- Full list on [Google Scholar](#).

Education

Master's in Energy Engineering

Sept 2025 – Present

University of Genoa, Savona Campus, Savona, Italy

Bachelor in Petroleum Engineering

Graduated 2023

NED University of Engineering and Technology, Karachi

- CGPA: 3.619 / 4.0

Certified Cloud Native Developer

Feb 2022 – Aug 2024

Presidential Initiative for AI & Computing (PIAIC) – Batch 36

Certifications

Google Data Analytics Professional Certificate – Google (Coursera)	Aug 2025
Google Advanced Data Analytics Professional Certificate – Google (Coursera)	Sept 2025
Google IT Automation with Python Professional Certificate – Google (Coursera)	Oct 2025
PCAP: Programming Essentials in Python – Cisco Networking Academy / OpenEDG Python Institute	Mar 2022
Excel Skills for Business: Essential, Intermediate I & II – Macquarie University (Coursera)	2021 – 2022
Lean Six Sigma Yellow Belt – Toosy Advisory Services	Jan 2021

Memberships & Achievements

- Society of Petroleum Engineers (SPE) – Executive Member, 2019 – Present
- American Chemical Society (ACS) – Community Associate, 2023 – Present
- University Representative, SPE ADIPEC University Programme – secured 2nd place among 13 international teams
- NED Alumni Association of Houston Scholarship recipient